K.R. MANGALAM UNIVERSITY

# Database Management Lab Sheet (ENCS205)

**Bachelor of Technology**

**in**

**Computer Science and Engineering**

*Submitted by*

***DHRUV-240101031***

B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

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## Objectives:

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

## Instructions

Open your preferred Database Management System (DBMS) tool.

1. Create the required tables: TEACHER, SALES, DEPARTMENTS, and EMPLOYEES.
2. Insert suitable records into each table.
3. Execute the SQL commands for transaction control, privilege control, grouping, sorting, and joins.
4. Observe the output after each operation such as ROLLBACK, COMMIT, SAVEPOINT, GRANT, and REVOKE.

**DDL Questions (Data Definition Language)**

1. Create the **TEACHER** table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).
2. Create the **SALES** table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).
3. Create the **DEPARTMENTS** table with the following attributes: (dept_no, dept_name, dept_location).
4. Create the **EMPLOYEES** table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

**DML Questions (Data Manipulation Language)**

**From TEACHER Table**

1. Insert five records into the **TEACHER** table.
2. Give an increment of 25% salary to teachers working in the Mathematics department.
3. Give an increment of 15% salary to teachers working in the Commerce department.

**From SALES Table**

1. Insert five records into the **SALES** table.
2. Calculate the total sales amount in each branch.
3. Calculate the average sales amount in each branch.

4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.
5. Display the Salesname and DOB of salesmen in alphabetical order of the month. **From**

**JOIN Operations**

1. Display employee details and department details where the department numbers are the same in both tables.
2. Display the employee name and department name using a LEFT OUTER JOIN.
3. Display the employee name and department name using a RIGHT OUTER JOIN.
4. Display the details of employees who draw a salary greater than the average salary.

**TCL Questions (Transaction Control Language)**

1. Perform the **ROLLBACK** command on the TEACHER table.
2. Perform the **COMMIT** command on the TEACHER table.
3. Write a query to implement a **SAVEPOINT**.

**DCL Questions (Data Control Language)**

1. Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.
2. Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.
3. Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.
4. Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

# ANSWERS:

## DDL Questions (Data Definition Language)

1. Create the TEACHER table with the following attributes:
(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```
CREATE DATABASE IF NOT EXISTS College_Lab4;

USE College_Lab4;

DROP TABLE IF EXISTS TEACHER;

CREATE TABLE TEACHER (
   Name VARCHAR(50),
   DeptNo INT,
   Date_of_Joining DATE,
   DeptName VARCHAR(50),
   Location VARCHAR(50),
   Salary DECIMAL(10,2)
);
```

2. Create the SALES table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).

```
DROP TABLE IF EXISTS SALES;

CREATE TABLE SALES (
   Sales_No INT PRIMARY KEY,
   Salesname VARCHAR(50),
   Branch VARCHAR(50),
   Salesamount DECIMAL(10,2),
   DOB DATE
);
```

3. Create the DEPARTMENTS table with the following attributes: (dept_no, dept_name, dept_location).

```
DROP TABLE IF EXISTS DEPARTMENTS;
CREATE TABLE DEPARTMENTS (
   dept_no INT PRIMARY KEY,
   dept_name VARCHAR(50),
   dept_location VARCHAR(50)
);
```

4. Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```
DROP TABLE IF EXISTS EMPLOYEES;

CREATE TABLE EMPLOYEES (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   emp_salary DECIMAL(10,2),
   dept_no INT
)
;
```

## DML – TEACHER Table Operations
## From TEACHER Table

1. Insert five records into the TEACHER table.

```
INSERT INTO TEACHER
(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary)
VALUES
('Anjali Sharma', 201, '2014-05-14', 'Mathematics', 'Jaipur', 52000),
('Karan Mehta', 202, '2016-03-18', 'Commerce', 'Ahmedabad', 47000),
('Sneha Verma', 203, '2018-11-21', 'Physics', 'Pune', 43000),
('Arjun Rao', 201, '2015-09-10', 'Mathematics', 'Lucknow', 56000),
('Priya Nair', 202, '2019-01-25', 'Commerce', 'Hyderabad', 49000);
```

2. Give an increment of 25% salary to teachers working in the Mathematics department.

```
UPDATE TEACHER
SET Salary = Salary * 1.25
WHERE DeptName = 'Mathematics';
```

3. Give an increment of 15% salary to teachers working in the Commerce department.

```
UPDATE TEACHER
SET Salary = Salary * 1.15
WHERE DeptName = 'Commerce';

SELECT * FROM TEACHER;
```

From SALES Table

Q1. Insert five records into the SALES table.

```
INSERT INTO SALES
(Sales_No, Salesname, Branch, Salesamount, DOB)
VALUES
(11, 'Ritika Singh', 'Jaipur', 18000, '1986-12-11'),
(12, 'Manish Kapoor', 'Pune', 24000, '1991-06-19'),
```

(13, 'Deepak Yadav', 'Jaipur', 21000, '1989-12-28'),
(14, 'Neha Joshi', 'Hyderabad', 26000, '1993-03-15'),
(15, 'Saurabh Jain', 'Pune', 30000, '1987-12-05');
Q2. Calculate the total sales amount in each branch.

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Q3. Calculate the average sales amount in each branch.

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Q4. Display the Salesname and DOB of salesmen born in the month of December with date format like 21-Dec-09.

```
SELECT Salesname,
DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

Q5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
SELECT Salesname,
DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
FROM SALES
ORDER BY MONTHNAME(DOB), DAY(DOB);
```

From JOIN Operations
(Insert sample data for DEPARTMENTS and EMPLOYEES first)

```
INSERT INTO DEPARTMENTS VALUES
(301, 'HR', 'Delhi'),
(302, 'Finance', 'Mumbai'),
(303, 'IT', 'Bangalore'),
(304, 'Marketing', 'Chennai');

INSERT INTO EMPLOYEES VALUES
(1, 'Rahul Sinha', 55000, 301),
(2, 'Meera Patel', 62000, 303),
(3, 'Amit Das', 48000, 302),
(4, 'Divya Reddy', 70000, 303),
(5, 'Kunal Roy', 45000, 304);
```

1. Display employee details and department details where the department numbers are the same in both tables.

```
SELECT e.emp_id, e.emp_name, e.emp_salary,
d.dept_name, d.dept_location
FROM EMPLOYEES e
JOIN DEPARTMENTS d
```

```
ON e.dept_no = d.dept_no;
```

2. Display the employee name and department name using a LEFT OUTER JOIN.

```
SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;
```

3. Display the employee name and department name using a RIGHT OUTER JOIN.

```
SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
RIGHT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;
```

4. Display the details of employees who draw a salary greater than the average salary.

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary >
(SELECT AVG(emp_salary) FROM EMPLOYEES);
```

TCL Questions (Transaction Control Language)

1. Perform the ROLLBACK command on the TEACHER table.

```
SET autocommit = 0;

INSERT INTO TEACHER
VALUES('Temporary User', 205, '2023-04-10', 'History', 'Indore', 38000);

ROLLBACK;
```

2. Perform the COMMIT command on the TEACHER table.

```
SET autocommit = 0;
INSERT INTO TEACHER
VALUES('Permanent User', 206, '2023-05-12', 'Chemistry', 'Surat', 46000);

COMMIT;
```

3. Write a query to implement a SAVEPOINT.

```
SET autocommit = 0;

INSERT INTO TEACHER
VALUES('SaveUser', 207, '2023-06-15', 'Commerce', 'Bhopal', 41000);

SAVEPOINT SP_One;

UPDATE TEACHER
SET Salary = Salary + 2000
WHERE Name = 'SaveUser';
```

```
ROLLBACK TO SP_One;
COMMIT;
```

## DCL Questions (Data Control Language)

1. Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT ALL PRIVILEGES
ON College_Lab4.EMPLOYEES
TO 'dept_user'@'localhost';
```

2. Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, UPDATE
ON College_Lab4.EMPLOYEES
TO 'dept_user'@'localhost';
```

3. Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL PRIVILEGES
ON College_Lab4.EMPLOYEES
FROM 'dept_user'@'localhost';
```

4. Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE UPDATE
ON College_Lab4.EMPLOYEES
FROM 'dept_user'@'localhost';
```